

Recent Adventures and Future Trends in AI

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Advancements: Claude 4.8, Grok 5, Antigravity 2.0

Claude Opus 4.8

- Anthropic's latest AI model
- Improved honesty compared to Claude Opus 4.7
- Several new features
- Improved agentic coding and terminal coding, improved multidisciplinary reasoning, improved agentic computer use, improved knowledge work, and improved agentic financial analysis.

	Opus 4.8	Opus 4.7	GPT-5.5	Gemini 3.1 Pro
Agentic coding SWE-Bench Pro	69.2%	64.3%	58.6%	54.2%
Agentic terminal coding Terminal-Bench 2.1	74.6%	66.1%	78.2%	70.3%
Multidisciplinary reasoning Humanity's Last Exam	49.8% no tools	46.9% no tools	41.4% no tools	44.4% no tools
	57.9% with tools	54.7% with tools	52.2% with tools	51.4% with tools
Agentic computer use OSWorld-Verified	83.4%	82.8%	78.7%	76.2%
Knowledge work GDPval-AA	1890	1753	1769	1314
Agentic financial analysis Finance Agent v2	53.9%	51.5%	51.8%	43.0%

Grok 5

- 6 trillion parameters trained on the world's first gigawatt-scale AI supercluster (Colossus 2)
- evolved multi-agent architecture from Grok 4.20
- Grok 5 will feature dynamic agent spawning, persistent memory across sessions, and cross-domain specialization
- real-time misinformation detection and fact-checking

Google Antigravity 2.0

- Three-component architecture: Desktop App + CLI + SDK instead of a single app
- Multiple AI agents work simultaneously on subtasks rather than a single assistant responding to prompts
- A 12x optimized version of Flash built specifically for Antigravity, outputting at 289 tok/s
- The Antigravity agent is available as a single API call on the Gemini API — it reasons, executes code, manages files, and browses the web in a secure Linux sandbox
- Usage is compute-based and refreshes every 5 hours instead of daily prompt limits
- Cross-platform: Available on macOS, Linux, and Windows

Future Trends in AI vs. Australia AI Framework and the EU AI Act

Artificial General Intelligence (AGI)

AGI is a theoretical stage of Machine learning where AI reaches or exceeds human intelligence when completing any task. AGI is the goal of AI to mimic human intelligence in technology.

Edge AI

Edge AI is the combination of AI with edge computing. Edge computing is data being processed locally on devices instead of the cloud. This allows for decisions to be made in real time, faster responses, and reduced dependency on wifi connection.

Neurosymbolic AI

Neurosymbolic AI is a combination of neural networks and symbolic AI. Neural networks struggle with explainability, rule-following, and transparency. Symbolic AI struggles with adaptability and learning from raw data. Therefore, combining them produces systems that are interpretable, adaptable, and trustworthy which is ideal for regulated, high-stakes fields. The goal is to enable human-like reasoning, contextual understanding, and explainable decision-making.

Quantum AI (QAI)

Quantum AI combines quantum computing with artificial intelligence to solve problems too complex or slow for classical computers. QAI to explore many possible solutions in parallel, enabling major speed-ups.

Australia AI Framework

Basically, Australia is shifting from voluntary ethics guidelines to a risk-based regulatory framework. Early approaches relied on soft law (2019 AI Ethics Principles, 2021 National AI Action Plan). The Australia National AI Plan 2025 marks a major shift toward structured, whole-of-economy governance. A new Australian Artificial Intelligence Safety Institute (AISII) will launch in early 2026 to provide independent safety testing and technical oversight.

EU AI Act

This was the 1st AI law. It bans high risk users, requires transparency, and fines $\leq 7\%$ global revenue.

Regulatory Coverage

AGI

Not directly regulated because it does not yet exist. Future AGI would require new global safety standards.

Neuro-Symbolic AI

Covered under general AI governance, especially explainability and accountability requirements.

Quantum AI

Falls under emerging AI and quantum research policies; not separately regulated.

Australia AI Framework

Covers all AI systems, including future AGI, through adaptable risk-based principles. Mandatory for government AI; voluntary standards for private sector.

EU AI Act

Covers all AI systems placed on the EU market. AGI would fall under General-Purpose AI with Systemic Risk, requiring strict testing, documentation, and oversight.

Ethical and Safety Priorities**AGI**

Alignment, control, existential risk, transparency, and preventing unintended autonomous behavior.

Neuro-Symbolic AI

Explainability, fairness, traceability, and reduced bias.

Quantum AI

Data integrity, algorithmic stability, and equitable access to quantum resources.

Australia AI Framework

Human oversight, transparency, privacy, fairness, and responsible deployment.

EU AI Act

Fundamental rights, safety, fairness, explainability, and preventing harmful or manipulative AI.

Governance Bodies**AGI**

Research labs (OpenAI, DeepMind, IBM), academic institutions.

Neuro-Symbolic AI

Universities, hybrid AI research centers, industry labs.

Quantum AI

Quantum computing leaders (IBM Q, Google Quantum AI, D-Wave).

Australia AI Framework

DISR, NAIC, OAIC, ACCC, eSafety Commissioner, and the Australian AI Safety Institute (AISI).

EU AI Act

European Commission, EU AI Office, national supervisory authorities.

Risk Management**AGI**

Conceptual risk management focused on preventing uncontrolled intelligence growth and ensuring alignment.

Neuro-Symbolic AI

Risk mitigation through explainability and rule-based reasoning.

Quantum AI

Risk tied to hardware instability, noise, and algorithmic unpredictability.

Australia AI Framework

Risk-based guardrails:

- Mandatory for government AI
- Voluntary standards (VAISS) for private sector

EU AI Act

Tiered risk classification:

- Unacceptable (prohibited)
- High risk (strict obligations)
- Limited risk (transparency)
- Minimal risk (no restrictions)

Transparency Requirements**AGI**

Future AGI would require strong transparency and interpretability, but no standards exist yet.

Neuro-Symbolic AI

Naturally supports transparency due to symbolic reasoning components.

Quantum AI

Transparency challenges due to quantum complexity; still emerging.

Australia AI Framework

Government AI must publish transparency statements; the private sector is encouraged to follow voluntary standards.

EU AI Act

Mandatory transparency for high-risk systems, general-purpose AI, and AI that interacts with humans (e.g., chatbots, deepfakes).

The EU AI Act is intentionally built as a living regulatory system. It adapts to fast-moving AI advances through these structures:

- The Act classifies AI into unacceptable, high, limited, and minimal risk categories.
- These categories can be updated without rewriting the entire law, allowing the EU to respond quickly to new AI capabilities.
- GPAI models (like large foundation models) have their own obligations, including documentation, copyright compliance, and training-data summaries.
- Systemic-risk GPAI models (e.g., trained with $>10^{25}$ FLOPs) must undergo adversarial testing, model evaluations, incident reporting, and cybersecurity protections. This is the EU's way of future-proofing the law for frontier-scale models.
- Until harmonized standards are published, GPAI providers can comply via codes of practice. These codes can be updated rapidly as AI evolves.
- A new EU AI Office monitors GPAI models, investigates systemic risks, and can update obligations as needed.
- Fast implementation timelines

Australia's approach is iterative, flexible, and risk-based, designed to evolve alongside AI progress through these structures put in place:

- Australia began with voluntary principles (AI Ethics Principles, VAISS). It is now transitioning toward mandatory guardrails for high-risk AI. This staged evolution allows rapid policy updates without waiting for full legislation.
- The 2024 Proposals Paper outlines 10 guardrails covering testing & evaluation, transparency, data governance, human oversight, accountability, and incident reporting. These guardrails are intentionally technology-neutral, so they can apply to future AI types.
- The Australian Artificial Intelligence Safety Institute (AISi) conducts independent safety testing, monitors emerging risks, and advises regulators. This gives Australia a technical capability to adapt rules as AI advances.
- Existing regulators (OAIC, ACCC, eSafety, TGA, APRA) enforce AI-related issues under existing laws. This means Australia can update sector-specific rules quickly without waiting for a single AI law.
- The National AI Plan 2025 emphasizes leveraging existing laws, adding new capabilities, and ongoing consultation.

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